min

Aptitude

Submitted

Analog

Submitted

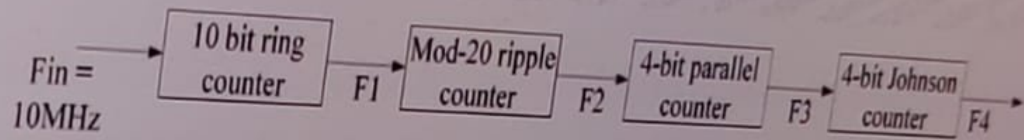
Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

1. The input frequency

‹ Previous Next ›

What is the value of F4 if the input frequency Fin in the given circuit is 10MHz?

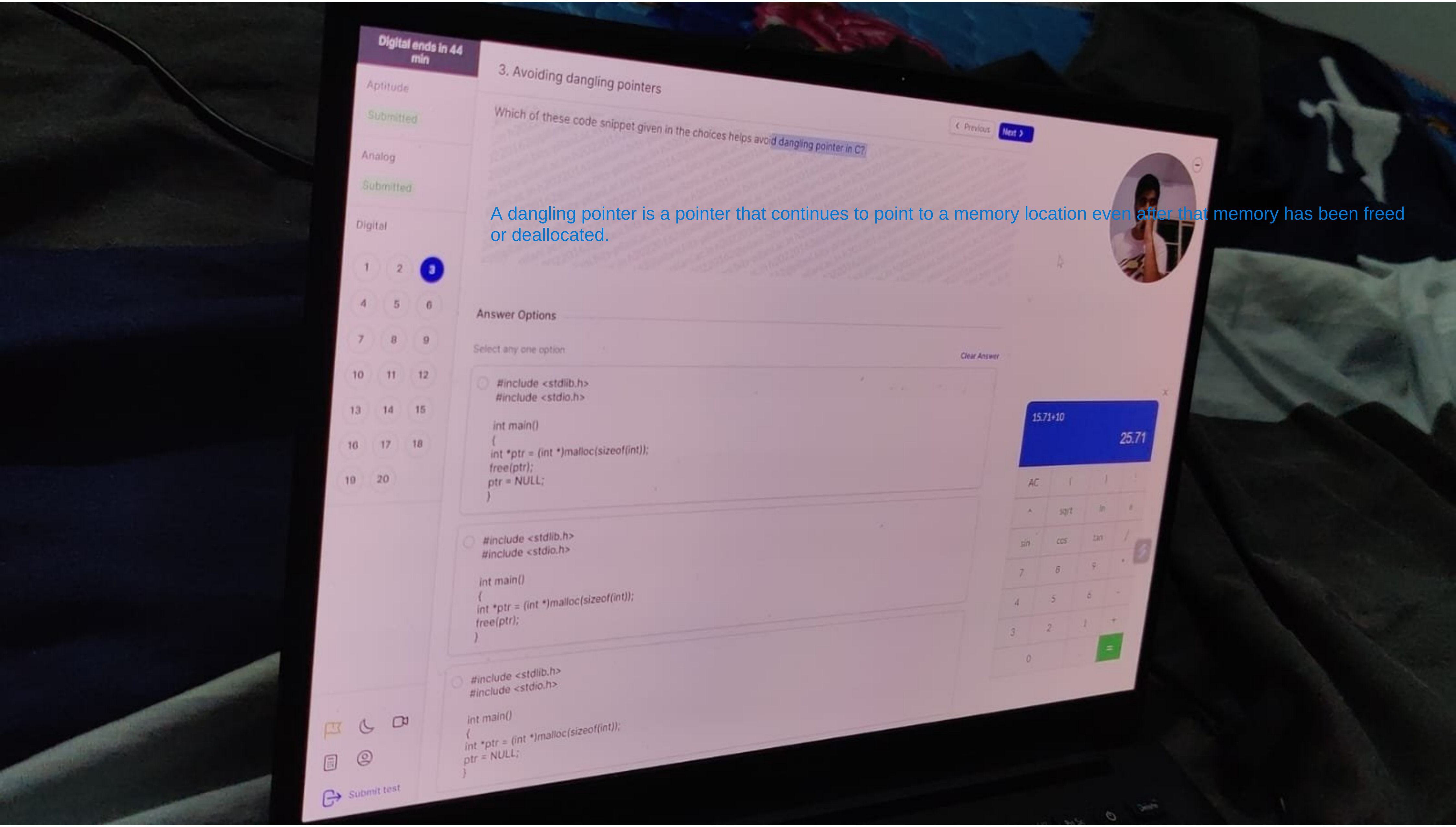


Answer Options

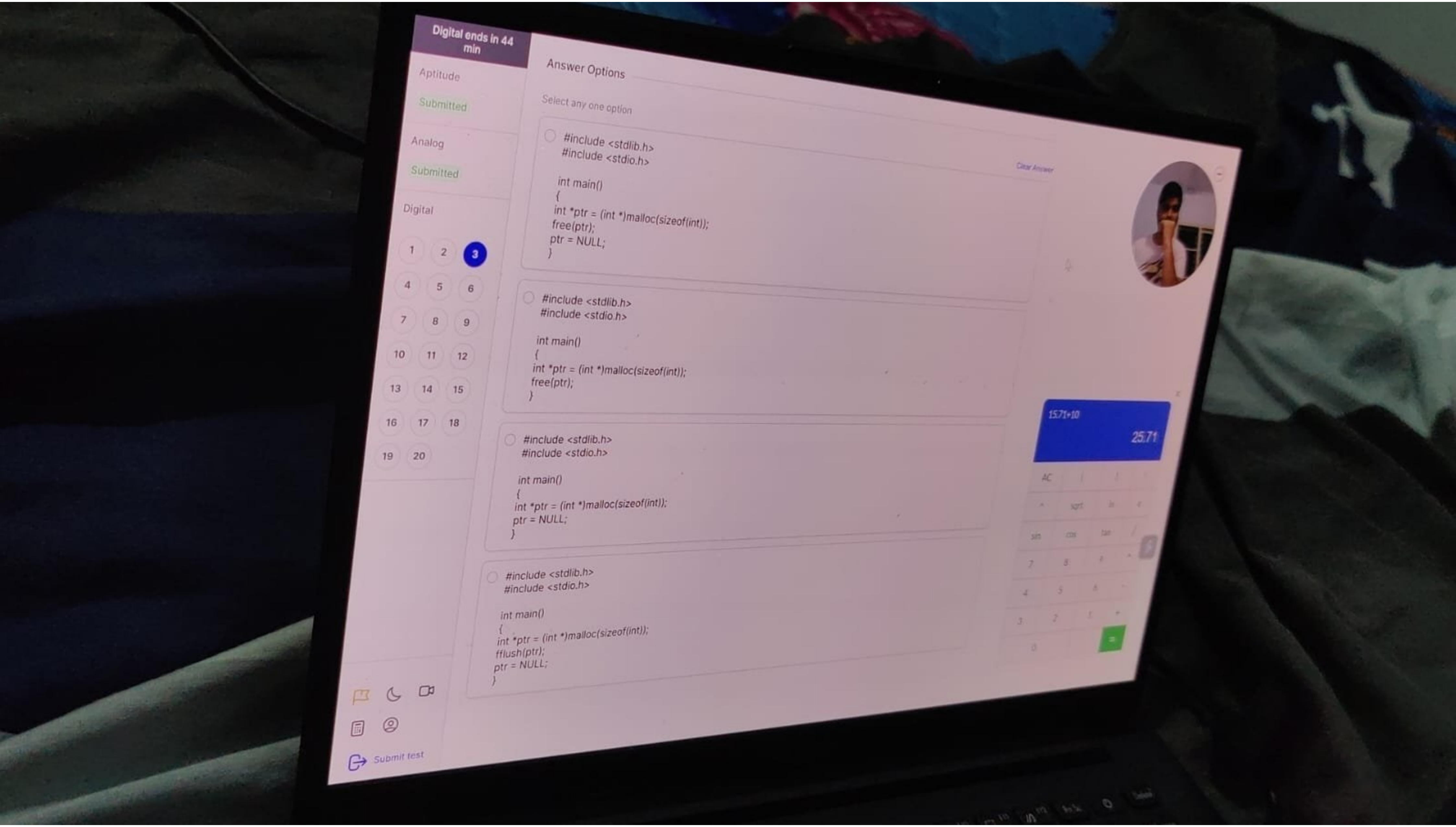
Select any one option

Clear Answer

- 390.625 Hz
- 402.5 Hz
- 250.25 Hz
- None of these



A dangling pointer is a pointer that continues to point to a memory location even after that memory has been freed or deallocated.



Digital ends in 44 min
Aptitude
Submitted
Analog
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Submit test
Answer Options
Select any one option
Clear Answer
#include <stdlib.h>
#include <stdio.h>
int main()
{
int *ptr = (int *)malloc(sizeof(int));
free(ptr);
ptr = NULL;
}
#include <stdlib.h>
#include <stdio.h>
int main()
{
int *ptr = (int *)malloc(sizeof(int));
free(ptr);
}
#include <stdlib.h>
#include <stdio.h>
int main()
{
int *ptr = (int *)malloc(sizeof(int));
ptr = NULL;
}
#include <stdlib.h>
#include <stdio.h>
int main()
{
int *ptr = (int *)malloc(sizeof(int));
fflush(ptr);
ptr = NULL;
}
15.71+10
25.71
AC
sqrt
sin
cos
tan

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

5. N bit numbers with base K

‹ Previous Next ›

You are asked to convert the following **N** bit number with base **K** to the corresponding number in base **K^2**.

> Xn-1 Xn-2 Xn-3 ... X1 X0

Which of these choices will help you achieve this?

Answer Options

Select any one option

Clear Answer

- ◯ Summation { (K^2)^ ((N-2)/2) x (KXm-1 + Xm-2) } where m belongs to N , 1 < m < N
- ◯ Summation { K^ ((N-2)/2) x (KXm-1 + Xm-2) } where m belongs to N , 1 < m < N
- ◯ Summation { (K^2)^ (N/2) x (KXm-1 + Xm-2) } where m belongs to N , 1 < m < N
- ◯ Summation { K^(N/2) x (KXm-1 + Xm-2) } where m belongs to N , 1 < m < N

15.71+10

25.71

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

4. Making a circuit hazard free

Consider the following boolean equation.

> F = summation (0, 4, 11, 13, 15) + d (1, 2, 3, 10)

How many redundant terms must be added to the minimized circuit in order to make it hazard free?

< Previous Next >

Answer Options

Select any one option

Clear Answer

- ( ) 0
- (•) 1
- ( ) 2
- ( ) 3

15.71+10

25.71

AC

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

7. Memory subsystem designing

Assume that you are working on a microcomputer system that requires large memory.

In the given context, how should the memory subsystem generally be designed?

Answer Options

Select any one option

- Static RAM
- Dynamic RAM
- Combination of static and dynamic RAM
- ROM

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

6. No. of cache sets

‹ Previous Next ›

You are working on a 16-bit microprocessor that uses a 2-way set-associative cache with a total capacity of 256 bytes. It is given that each cache line contains 8 bytes.

How many cache sets are there in the cache?

Answer Options

Select any one option

Clear Answer

- 8
- 16
- 32
- 64

15.71+10

25.71

AC

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

9. Application of CMOS devices

Which of the following statements about CMOS devices is/are valid?

Answer Options

Select any one option

- The devices should not be inserted into circuits with the power on
- All tools, test equipment and metal workbenches should be tied to earth ground
- The devices should be stored and shipped in antistatic tubes or conductive foam
- All of these

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

10. Address Lines calculation

‹ Previous Next ›

You want to address a memory location out of N memory locations.

How many address lines would you require to do so?

Answer Options

Select any one option

Clear Answer

- (selected) log N (to the base 2)
- log N (to the base 10)
- log N (to the base e)
- log (2N) (to the base e)

15.71+10

25.71

AC

Digital ends in 42 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

11. A pulse train

‹ Previous Next ›

A pulse train with a frequency of 1 MHz is counted using a modulo - 1024 ripple counter built with J-K flip flops.

What will be the maximum permissible propogation delay per flip flop stage for the proper operation of the couter?

Answer Options

Clear Answer

Select any one option

- 100 nsec
- 10 nsec
- 1 nsec
- 50 nsec

15.71+10

25.71

AC ( )

sqrt

sin cos tan

7 8 9

4 5 6

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

12. Working with arrays

‹ Previous Next ›

Predict the output of the following C code snippet.

```
bool compare (const void * a, const void * b)
{
return ( *(int*)a == *(int*)b ); //L1
}
int search(void *arr, int arr_size, int ele_size, void *x,
bool compare (const void * , const void *)) //L2
{
char *ptr = (char *)arr;
int i;
for (i=0; i<arr_size; i++)
if (compare(ptr + i*ele_size, x)) //L3
return i;
return -1;
}

int main()
{
int arr[] = {2, 5, 7, 90, 70};
int n = sizeof(arr)/sizeof(arr[0]);
int x = 7;
printf ("The index is %d ", search(arr, n, sizeof(int), &x, compare));
return 0;
}
```

15.71+10

25.71

Answer Options Clear Answer

Select any one option

- The index is 2
- The code will throw an error at L1 and L2
- The code will throw an error at L2
- The code will throw an error at L2 and L3

Aptitude ends in 30 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
17 18
20

30 min

ftware ›

1. The oldest sister

‹ Previous Next ›

There are four sisters P, S, R, and T. P is younger than S, and R is older than T. Which of the given data will allow you to find out the oldest among the four sisters?

Statement 1: R is older than P
Statement 2: S is older than R
Statement 3: T is the youngest of all

Answer Options

Select any one option

Clear Answer

- Only Statement 1
- Only Statement 2
- Only Statement 3
- Both Statement 1 and Statement 2

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

oftware

arts in 29 min

to ware

2. Forming strings

< Previous Next >

The letters A, A, B, M, and T can be used to form 5-letter strings such as MABAT or MAABT. You have to use these letters to form 5-letter strings such that two occurrences of the letter A are separated by at least one letter.

In the given context, how many strings can be formed?

Answer Options

Select any one option

Clear Answer

- 24
- 36
- 32
- 34

5! / 2! - 4! * 2! / 2!

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
20

3. Determine value of X

< Previous Next >

What can be used in place of **X** to complete the given series?

17, 24, 33, 44, **X**

Answer Options

Select any one option

Clear Answer

- 55
- 57
- 66
- 65

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
14 15
17 18
20

29 min

4. Deliver packages

‹ Previous Next ›

Truck Driver A and Truck Driver B, work together to ship certain packages in 7.5 days. A works alone and delivers half the job. If B takes over and delivers the remaining half alone, they will deliver all the packages in 20 days.

How long will B alone take to deliver the packages if A is more efficient than B?

Answer Options

Select any one option

Clear Answer

- 20
- 30 ✓
- 22
- 28

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
11 12
14 15
17 18

5. Vehicle tracking devices

< Previous Next >

Mr. Park has two drivers A and B, who drive two different cars. He has installed a vehicle tracking device in both vehicles. On a particular day, Driver B left at 10:00 a.m. heading south, and driver A left at 2:00 p.m. heading north.

Later that day, at a particular time, data was retrieved from both the tracking devices. It showed that, up to that time, A had averaged 60 miles per hour, and B had averaged 40 miles per hour. It also showed that A and B had driven a combined total of 460 miles.

At what time was the data retrieved from the vehicle tracking devices?

Answer Options

Select any one option

Clear Answer

- 5:00 PM
- 4:00 PM
- 3:00 PM
- 2:00 PM

Aptitude ends in
28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

ftware

rts in 28 min

to
ware

nit

6. The rate of the current

Previous Next

A fisherman can row 20 km upstream and 40 km downstream in 6 hours. He can also row 25 km upstream and 35 km downstream in 7 hours.

In the given context, what is the **rate of the current?**

Answer Options

Select any one option

Clear Answer

- 12.5 km/hr
- 13.5 km/hr
- 14.5 km/hr
- 15.5 km/hr
- Cannot be determined

ptitude ends in 28 min

titude

2 3

5 6

8 9

11 12

4 15

18

min

7. Trains A and B

Two trains A and B are running on parallel tracks in the same direction. The length of A is 65 m and it is running at a speed of 34km/hr. The length of B is 70 m and it is running at a speed of 25 km/hr.

How long will it take for both the trains to pass each other completely from the moment they met?

Answer Options

Select any one option

Clear Answer

- 54 seconds
- 56 seconds
- 50 seconds
- None of these

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
3 14 15
17 18
20

are

in 28 min

e

8. Find Julian's age

Previous Next

Rob was introduced to his colleague Julian during the founder's day celebration at his company in 2016. Four years later, Rob runs into Julian during the 8th founder's day celebration on 29th February 2020.

If Julian remarks that he is only four years younger than the company's age, what is Julian's current age?

Note: *It is given that founder's day is celebrated only on 29th February.*

Answer Options

Select any one option

Clear Answer

- 28
- 36
- 34
- 30

Aptitude ends in 27 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
14 15
17 18
20

27 min

9. Zoey's age

‹ Previous | Next ›

Which of these statements is sufficient to determine the age of Zoey in 2050?

Statement I: When Zoey was born her father was 32 years old, who is 4 years older than Zoey's mother.
Statement II: Zoey's mother was born in the first leap year that came after 1970.

Answer Options

Select any one option

Clear Answer

- Statement I ALONE is sufficient, but Statement II ALONE is not sufficient. the question
- Statement II ALONE is sufficient, but Statement I ALONE is not sufficient. the question
- BOTH statements TOGETHER are sufficient, but EITHER statement ALONE is insufficient.
- EACH statement ALONE is sufficient.

Aptitude ends in 27 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
14 15
17 18
20

27 min

10. A parking lot

Previous Next

A parking lot has several vehicles facing North. In a particular row, V1 is 20th from the left of V2, who is 27th from the right end. Another vehicle V3 is 19th from the left end and 11th from the right of V1.

How many vehicles are there in this row?

Answer Options

Select any one option

Clear Answer

- 44
- 54
- 50
- 40

Aptitude ends in 27 min

11. Capacity of the pool

Aptitude

1 2 3
4 5 6
7 8 9
0 11 12
14 15
17 18
20

You are asked to use two pipes P1 and P2 to fill a pool that has another pipe P3 at the bottom of it. It is given that pipes P1 and P2 can fill the tank in 10 and 20 minutes, respectively and pipe L3 can empty 12 gallons per minute from the pool.

What is the **capacity of the pool** if all three pipes working together can fill the tank in 30 minutes?

Note: *Choose the option that coincides with the nearest integer.*

Answer Options

Select any one option

Clear Answer

- 100 gallons
- 150 gallons
- 103 gallons
- 125 gallons

re

27 min

Aptitude ends in 26 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

ftware

rts in 26 min

to Software >

it test

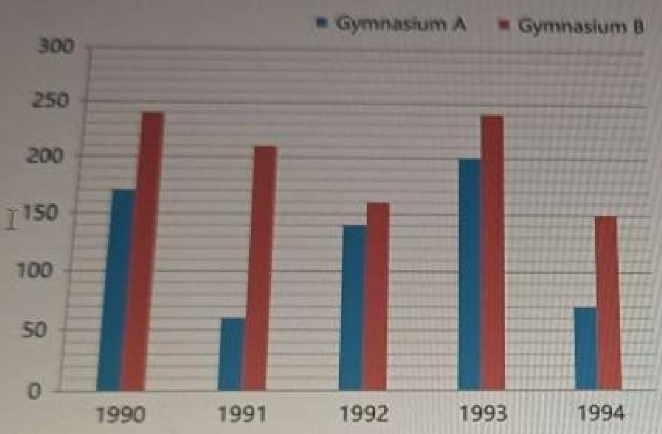
12. Gymnasiums A and B

< Previous Next >

The given bar graph shows the total number of members enrolled in gymnasiums A and B from 1990 to 1994.

For both A and B, it was observed that in the year 1995, there was a **30% increase** in the total number of members enrolled in comparison to the enrollment in 1994.

Find the total number of members enrolled in 1995.



Answer Options

Select any one option

Clear Answer

- 282
- 296
- 292
- 286

min

Aptitude

1 2 3

4 5 6

8 9

11 12

14 15

17 18

20

min

vare >

13. A conference

Previous Next >

A conference for budding entrepreneurs is being conducted at an engineering college. A total of 200 students of varying age groups are attending this conference.

In the given scenario, which of these statements will help you determine if there are any students in the conference who are of the same age and from the same engineering college?

Statement 1: The range of ages of the students is 18 to 22, inclusive.
Statement 2: Students from around 15 engineering colleges are attending this conference.

Answer Options

Select any one option

Clear Answer

- Statement 1 alone is sufficient
- Statement 2 alone is sufficient
- Both Statements together is sufficient
- Either Statement 1 or Statement 2 is sufficient

Aptitude ends in 25 min

ptitude

1 2 3

5 6

8 9

11 12

14 15

17 18

0

min

/are >

14. An online learning platform

< Previous Next >

An online learning platform provided less amenities and offers to people who had a monthly membership. Market research is conducted and it is found that the learning platform users are willing to pay a higher price for better amenities and offers. It has also been found that the higher price would cover the cost for these improvements. However, the implementation of this scheme has not been decided yet even though the learning platform is trying to make more profits.

Which of these choices, if true, would most help to explain the learning platform's decision in light of its objectives?

Answer Options

Select any one option

Clear Answer

- None of the learning platform's competitors offer significantly better amenities and offers to its monthly membership u sers.
- The number of people who would be willing to pay the high fares to be a 6 month and 12 month user would decrease.
- A few of the learning platform's users are satisfied with the service they receive, given the low price they pay.
- Very few users avoid using the learning platform because of less amenities and offers given by the platform to its mont hly membership users.

Aptitude ends in 25 min

Aptitude

1 2 3

4 5 6

8 9

11 12

14 15

17 18

15. A gourmet bakery

‹ Previous Next ›

A gourmet bakery decides to launch three limited desserts B1, B2 and B3. In 2020, the bakery launched the desserts in the ratio 2:3:4, respectively. In 2021, the bakery removed **X** items from each of the three types of desserts that were launched in 2020 such that the new ratio is 3:5:7.

In 2022, the bakery decides to remove 10 items from each of three types of desserts produced in 2021 such that the new ratio is 1:2:3.

How many items did the bakery remove in total in the year 2021?

Answer Options

Select any one option

Clear Answer

- 35
- 30
- 40
- 45

min

are ›

Aptitude ends in 25 min

ptitude

2 3

5 6

8 9

11 12

15

18

16. Jill and Adam in an airport

Previous Next

Two people, Jill and Adam are running towards each other on a moving walkway of length 4500m in an airport. Jill is running at a speed of 10 m/s and Adam is running at a speed of 5 m/s.

What is the time taken for Jill and Adam to meet if the speed of the moving walkway is 40 m/s?

Answer Options

Select any one option

Clear Answer

- 55 sec
- 53 sec
- 62 sec
- 65 sec

Aptitude ends in 25 min

Aptitude

1 2 3
4 5 6
7 8 9
11 12
14 15
17 18
20

min

vare >

17. New clock in a library

Previous Next >

On a particular day, a library installed a new clock which gains time uniformly. The librarian observes the clock on a Monday morning and notices that the clock is 10 mins slow at 10:00 AM.

She then observes the clock again on the following Monday and notices that the clock is 10 mins 50 seconds fast at 10:00 PM.

Analyse the given scenario and determine when this clock will show the correct time.

Note: *Select the accurate time in hours and minutes.*

Answer Options

Select any one option

Clear Answer

- 12:25 pm on Thursday
- 12:25 am on Thursday
- 10:00 pm on Thursday
- 10:25 pm on Wednesday

Aptitude ends in 25 min

Aptitude

1 2 3
4 5 6
7 8 9
11 12
14 15
17 18
20

5 min
vare ›

18. The alphanumeric series

‹ Previous Next ›

Which of these choices comes next in the given alphanumeric series?

KM10, PR15, UW20, XZ23, ...

Answer Options

Select any one option

Clear Answer

- CE2
- EG22
- CE22
- EG2

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

Submit test

1. Working with pointers

Predict the output of the following code snippet.

```
#include <stdio.h>

int main(void) {
int numbers[5];
int * p;
p = numbers;
*p = 10;
p++;
*p = 20;
p = &numbers[2];
*p = 30;
p = numbers + 3;
*p = 40;
p = numbers;
*(p + 4) = 50;
for (int n = 0; n < 5; n++)
printf("%d,",numbers[n]);
return 0;
}
```

Next >

Answer Options

Select any one option

Clear Answer

- 10,20,30,40,50,
- 1020304050
- Compile time error
- Runtime error

min

Aptitude

Submitted

Software

1 2 3

4 5 6

8 9

2. Keeping the CPU and the I/O devices busy

< Previous | Next >

Which of these techniques can be used to keep both the CPU and the I/O devices busy simultaneously in an operating system?

Answer Options

Select any one option

Clear Answer

- Preemptive scheduling
- Multiprogramming
- Spooling
- All of the given options

Software ends in 30 min

Aptitude

Submitted

Software

1 2 3

5 6

8 9

3. The UART communication protocol

‹ Previous Next ›

You are using the following 8051 code to send a char using the UART communication protocol.

```
void UART_TxChar(char ch)
{
SBUF = ch;
while(TI==0);
}
```

What is missing in the code?

Answer Options

Select any one option

Clear Answer

- The TI flag is not cleared
- The TI flag is initialised
- Both Choice 1 and 2
- Neither Choice 1 nor 2

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

4. Working with structures

< Previous Next >

What output will the following code snippet produce when it is run on a 64-bit machine?

```
struct{
float angle, radius;
} *vector[10];

printf("%d\n", sizeof(vector));
```

Answer Options

Select any one option

Clear Answer

- 20
- 32
- 64
- 80

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

5. Managing deadlocks

Previous Next

Read the following statements about deadlocks carefully and choose the correct option.

A: Circular wait is a necessary condition for deadlocks to occur.
B: Circular wait is a sufficient condition for deadlocks to occur.

Answer Options

Select any one option

Clear Answer

- A and B are both true.
- A is true, B is false.
- A is false, B is true.
- A and B are both false.

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

6. Identify dependency between instructions

< Previous Next >

You come across two instructions i and j where j tries to write to a register before i reads that register.

In the given context, what type of dependency exists between i and j?

Note: *The original ordering must be preserved to ensure that i reads the correct value.*

Answer Options

Select any one option

Clear Answer

- Order dependence
- Anti-dependence
- Output dependence
- True dependency

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

7. Understanding circular linked lists

‹ Previous Next ›

You are running the following function on a circular linked list. What will be the result of executing the function?

```
void sample()
{
struct node *ptr,*temp;
int item;
ptr = (struct node *)malloc(sizeof(struct node));
if(ptr == NULL)
{
printf("\nOVERFLOW");
}
else
{
scanf("%d",&item);
ptr → data = item;
if(head == NULL)
{
head = ptr;
ptr → next = head;
}
else
{
temp = head;
while(temp→next != head)
temp = temp→next;
ptr→next = head;
temp → next = ptr;
head = ptr;
}
}
}
```

Note: *head here represents the first node of the linked list.*

Answer Options

Select any one option

omit test

It inserts a node at the beginning of the list

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

7. Understanding circular linked lists

```
scanf("%d",&item);
ptr → data = item;
if(head == NULL)
{
head = ptr;
ptr → next = head;
}
else
{
temp = head;
while(temp→next != head)
temp = temp→next;
ptr→next = head;
temp → next = ptr;
head = ptr;
}
}
}
```

Note: *head here represents the first node of the linked list.*

Answer Options

Clear Answ

Select any one option

- It inserts a node at the beginning of the list.
- It inserts a node at the end of the list.
- It searches for a particular node in the list.
- None of these

nit test

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

8. Implementing an algorithm

< Previous Next >

Which of the following algorithms is being implemented below?

1. Sort all the edges in non-decreasing order of their weight.
2.Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If cycle is not formed, include this edge. Else, discard it.
3.Repeat step#2 until there are (V-1) edges in the spanning tree.

Answer Options

Select any one option

Clear Answer

- Prim's algorithm
- Kruskal's Algorithm
- Shortest path algorithm
- Depth first search algorithm

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

9. Sorting an array

‹ Previous Next ›

Consider the following algorithm to sort an array.

```
function MergeSort(A, n) //A[1:n] is an array of n integers
 if (n==1)
 return (A)
 else if (n==2)
 return ( [min(A[1],A[2]), max(A[1],A[2]])
 else begin
 B = MergeSort(A[1:floor(n/2)], floor(n/2))
 C = MergeSort(A[ceil(n/2)],n], ceil (n/2))
 return (Merge(B,C))
 end
endfunction
```

Analyse the algorithm and determine the complexity of the function **MergerSort()** if the complexity of **Merge()** is **O(n)**.

Answer Options

Select any one option

Clear Answer

- O(n^2)
- O(n^3)
- O(n)
- None of these

Software ends in 29 min

Aptitude

Submitted

Software

1 2 3

4 5 6

7 8 9

10

10. A weird fruit

‹ Previous Next ›

You have a weird fruit with you that you can either sell as it is or you can divide it magically into 3 parts 1/2, 1/3 , 1/4 and sell them. It is given that the division performed is an integer division and initially the weight of the fruit is **n** units.

You have to determine the maximum weight of the fruit that you can obtain from the magical division procedure. You are given the following pseudocode for performing this operation.

```
Solve(n):
if(n==0):
return 0

return X
```

Analyse the code and choose the correct option that fills the blank **X**.

Note:
Example: n=2, if you divide it you will get 1,0,0 weights i.e 1 unit, so max is 2 unit of weight
Example: n=12, if you divide it you will get 6,4,3 so total is 13 (you don't divide them further).

Answer Options

Select any one option

Clear Answer

- X:max(n,solve(n/2)+solve(n/3)+solve(n/4))
- X: solve(n/2)+solve(n/3)+solve(n/4)
- X:n
- X:max(n+3,solve(n/2)*solve(n/3)*solve(n/4))